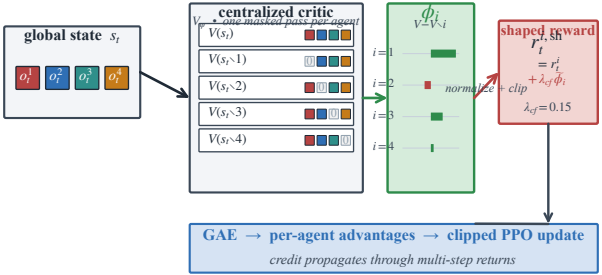


a Leave-one-out counterfactual credit



Cost: N extra critic passes per timestep • reuses centralized V_π

b Vignette • shared r_t vs. counterfactual credit

Both agents reach their goals — same shared $r_t = +1.0$			
			
view	yields Agent A	passes through Agent B	credit?
shared r_t	+1.0	+1.0	ambiguous
ϕ_i	+0.72	+0.08	A enables B
$r_t^{i, \text{sh}}$	+1.11	+1.01	yielding rewarded

$r_t^{i, \text{sh}} = r_t^i + \lambda_{cf} \bar{\phi}_i$ with $\lambda_{cf} = 0.15$, clip ± 2.5